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DEVELOPING A HEALTH AND PREDICTIVE MAINTENANCE SYSTEM FOR ELECTRIC MOTORS

Predictive Maintenance via Real-Time Sensor Fusion

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ENGG8201 INTERNET OF THINGS SYSTEMS DESIGN

PRESENTATION HIGHLIGHTS

FOCUS AREAS

- Motivation
- Project goal
- overview of the system
- IoT node
- Communication protocol
- API and Framework
- Demonstration of IoT Node
- Data collected from the IoT node
- Machine Learning
- Challenges & Advantages
- Proposed future work
- Conclusion

MOTIVATION: WHY MOTOR HEALTH MONITORING?

44%
of unplanned downtime caused by equipment failure

- Electric motors = the most widely used industrial equipment
- Failures cause production stoppages & major costs
- Traditional preventive maintenance = unnecessary downtime
- Industry 4.0: IoT + ML enables condition-based predictive maintenance

REACTIVE	PREVENTIVE	PREDICTIVE ✓
Repair after failure	Scheduled service	Condition-based IoT

Typical Downtime per unexpected failure

42-75 Hours

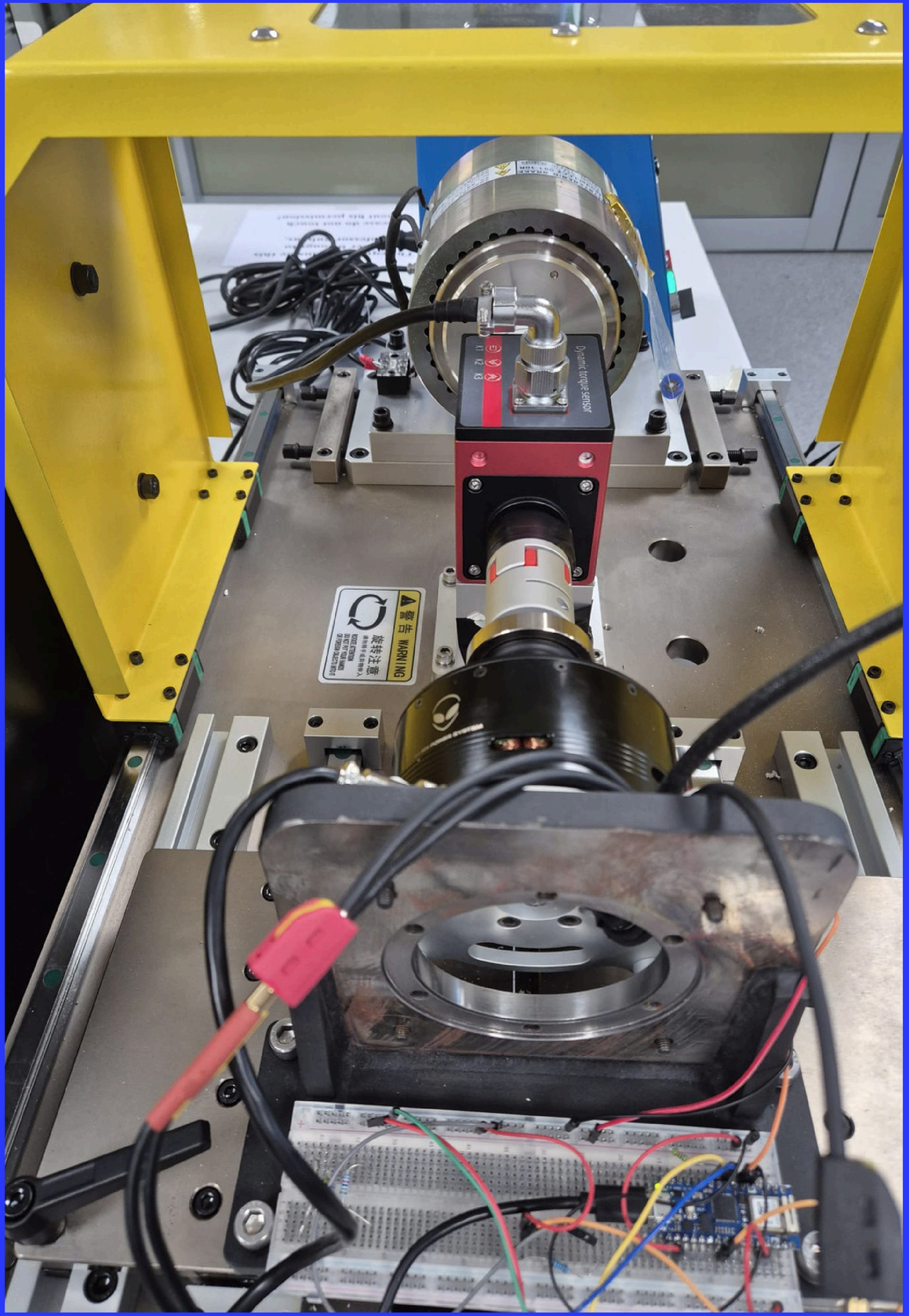
Cost rate during unplanned shutdown

$10k - $50k/H

Combined cost per incident

Up to $3.75 M

IMPACT ON UNPLANNED SHUTDOWN



IOT: THE HERE AND NOW

AVAILABLE TECHNOLOGY

SENSOR INTERFACE

- SHT11 - Digital (2-wire serial) - Temperature
- Shunt Resistor - Analog - Current
- A3144 Hall - Digital - RPM

ARDUINO

- SAMD21 Cortex®-M0+ 32bit low power ARM MCU
- Nina W102 uBlox module
- LSM6DS3 onboard accel. - I2C - Vibration

HOW DO WE COMMUNICATE?

INCREASED CONNECTIVITY LEADS TO HIGHER-LEVEL SOLUTIONS

The IoT Simplified

Communication Protocol and Data flow

WiFi Choosen Becasue

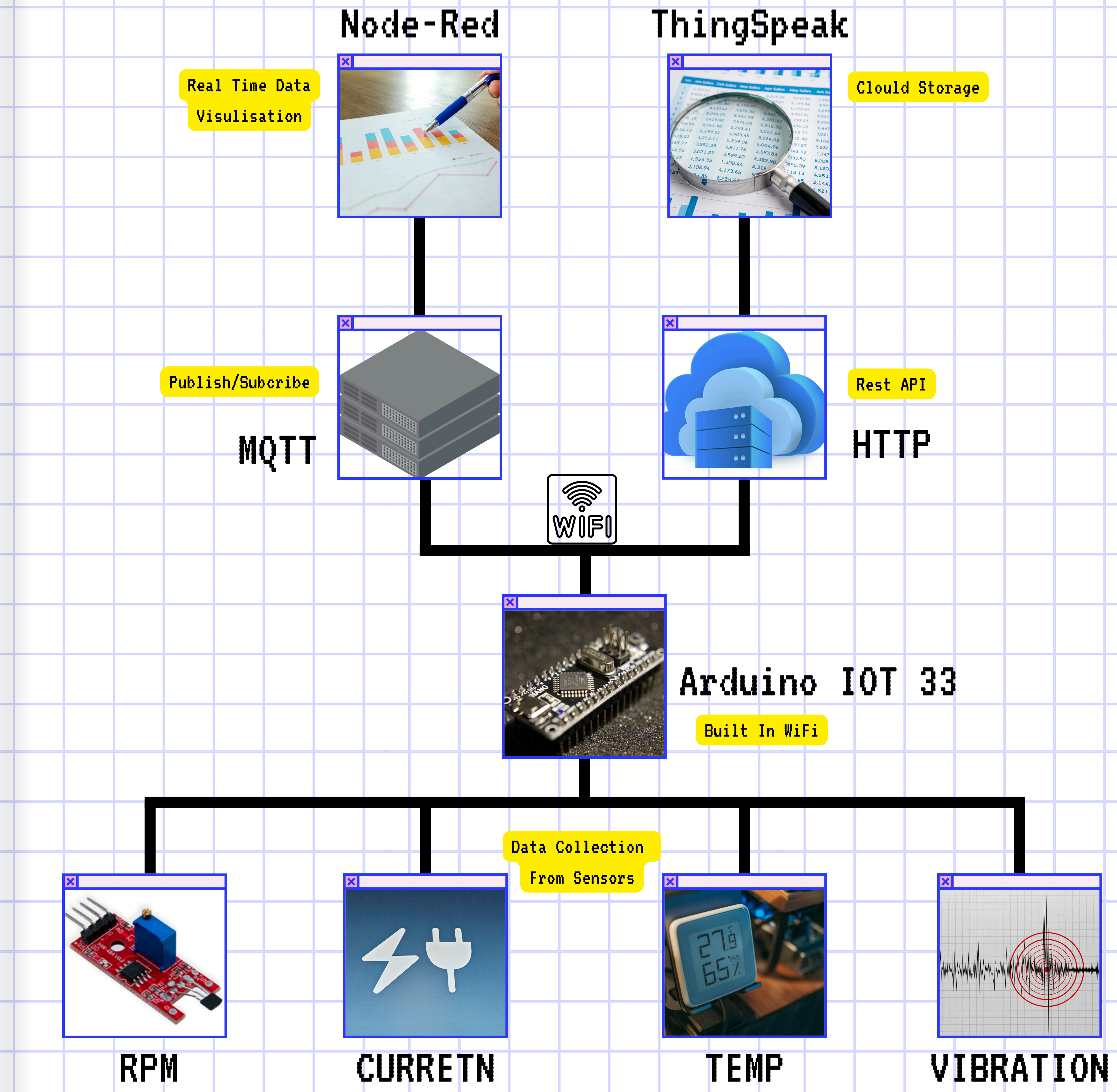
- Built into Nano 33 IoT · Direct internet
- High data rate for multi-sensor real-time streaming
- Industrial plants have WiFi infrastructure

MQTT Choosen Becasue

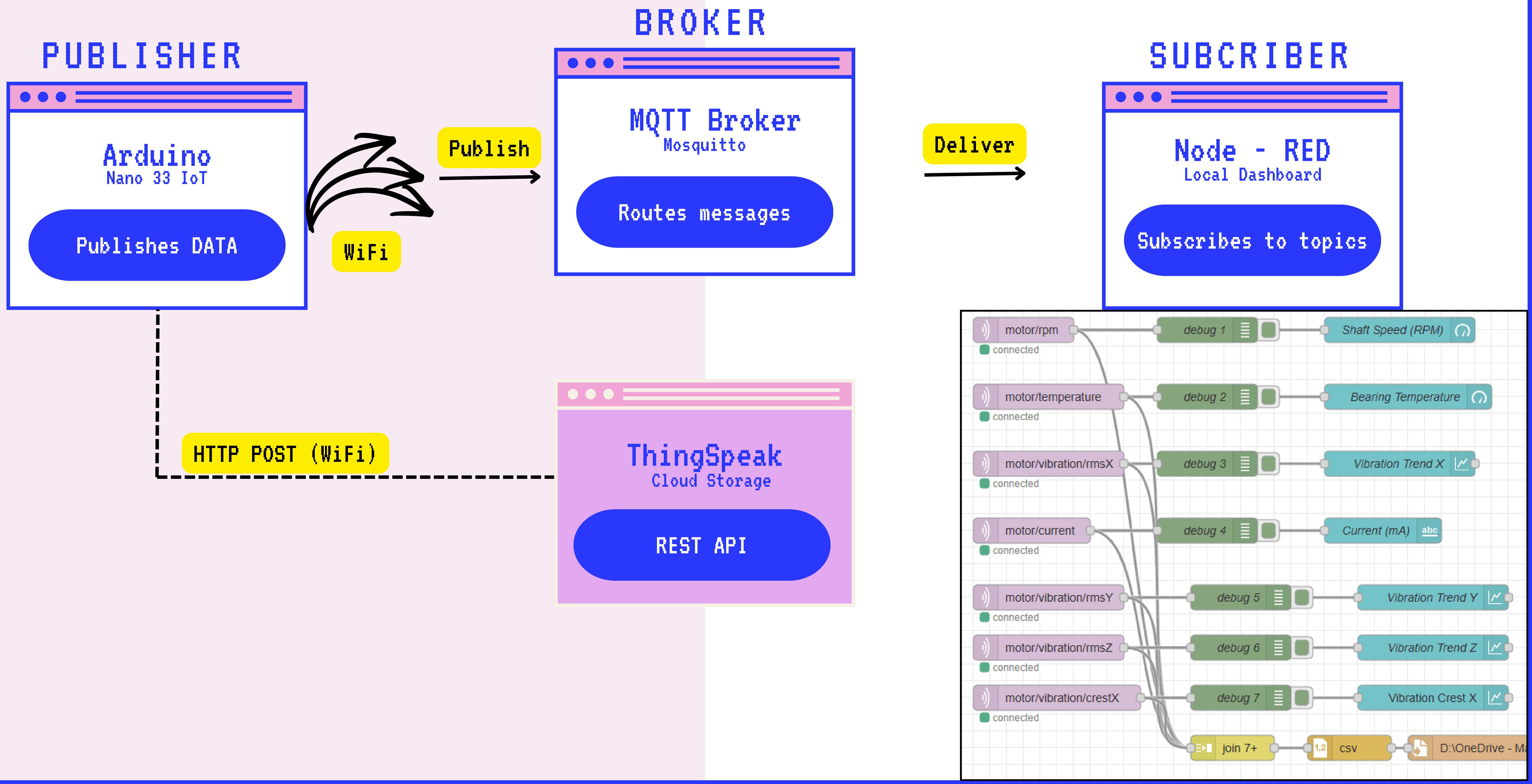
- Publish-subscribe model · Low bandwidth overhead
- Low latency · Scales to multiple devices
- Native Node-RED support

ThingSpeak Choosen Becasue

- Long-term dataset logging · Remote access
- MATLAB analytics for future ML integration



API & DASHBOARD



COLLECTED DATASETS

Test Conditions

Low Speed - 500 to 600 RPM - Healthy

High Speed - 1030 RPM - Healthy

High Speed - 1030 RPM - Loose Screw

High Speed - 1030 RPM - Shaft Misalignment

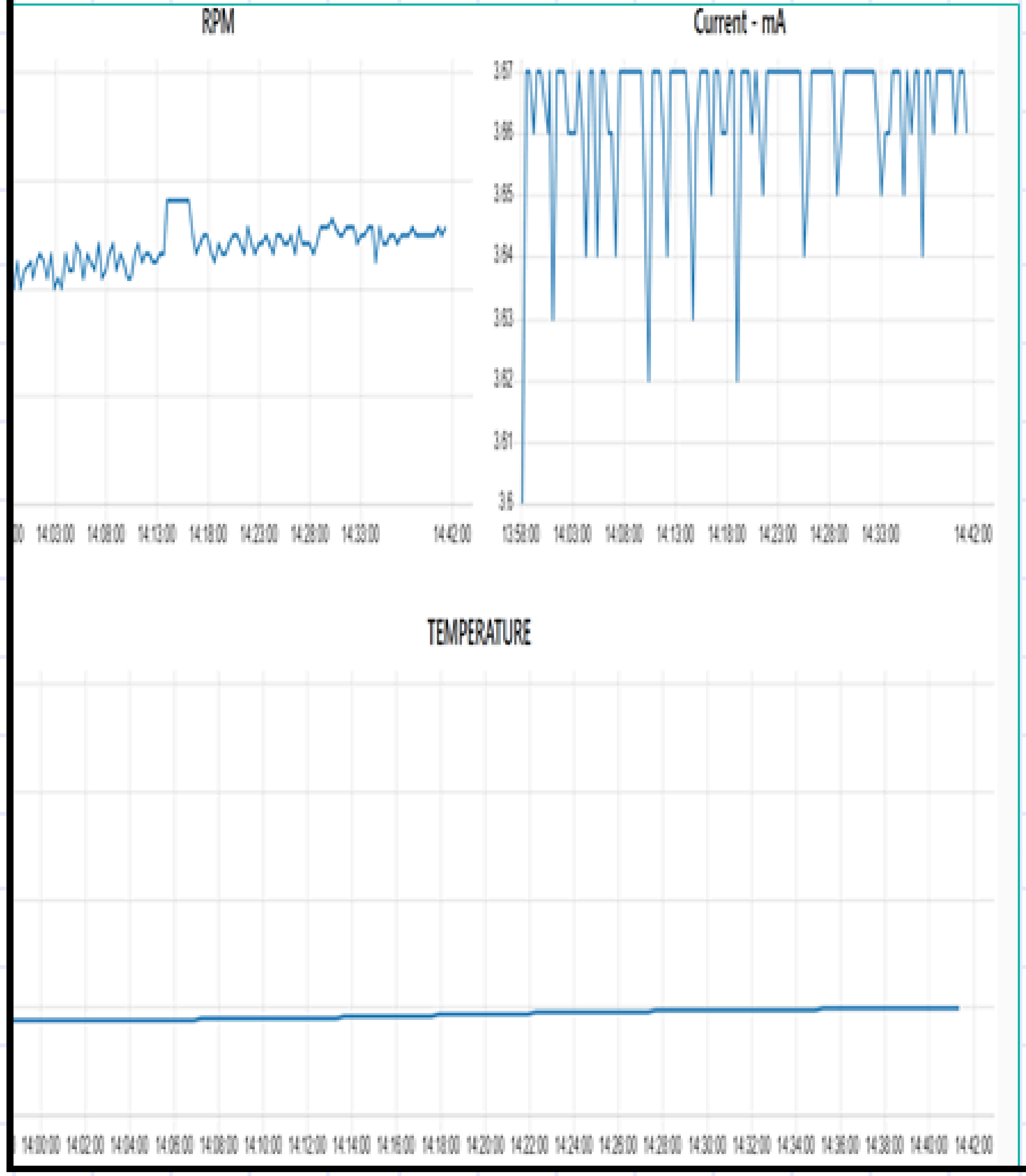
KEY FINDINGS

DATASET — NORMAL OPERATION (~500–600 RPM)

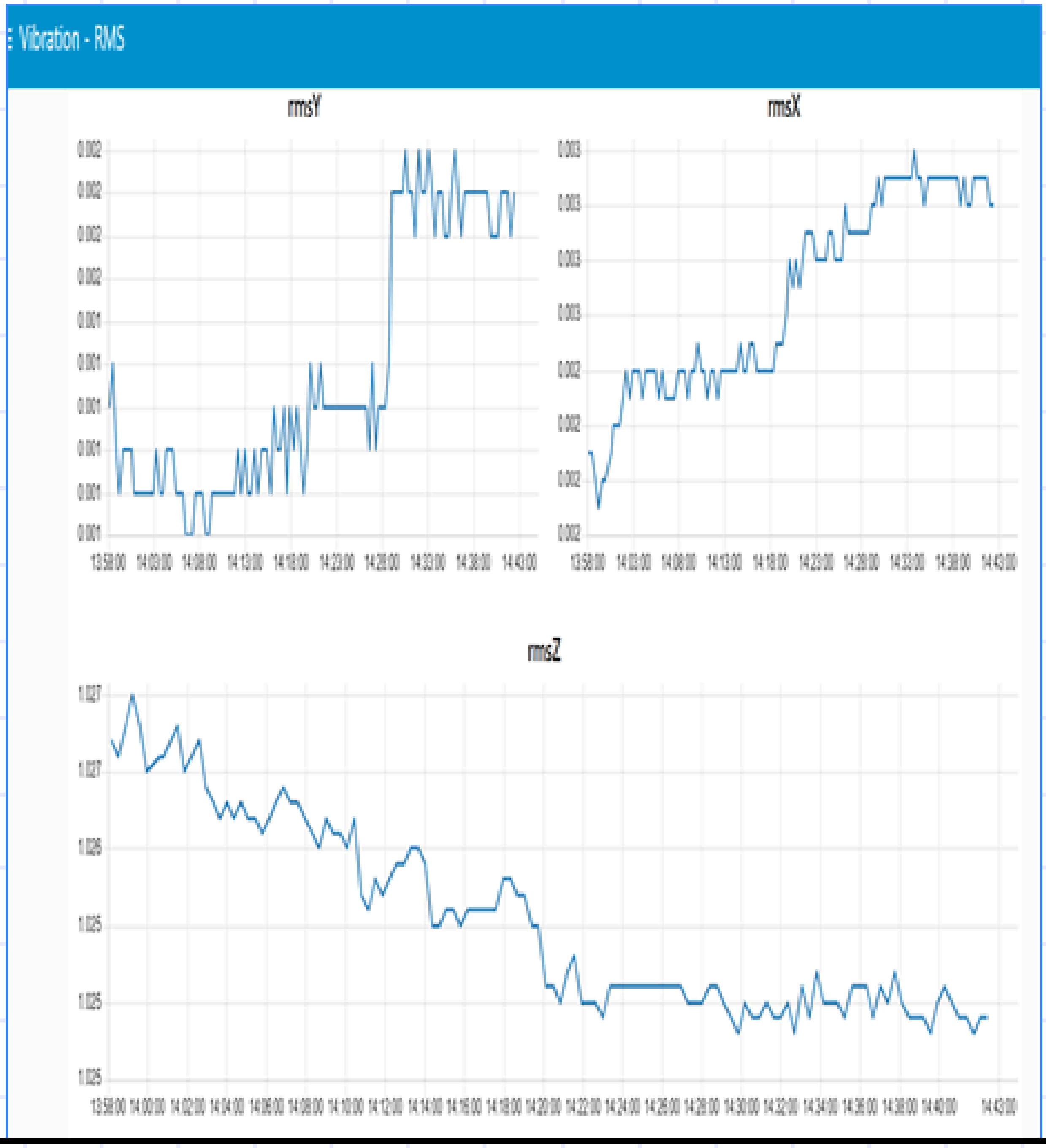
✓ NORMAL

Baseline healthy condition · Temp: 21–24°C · RPM: ~600 · Current: ~3.66A

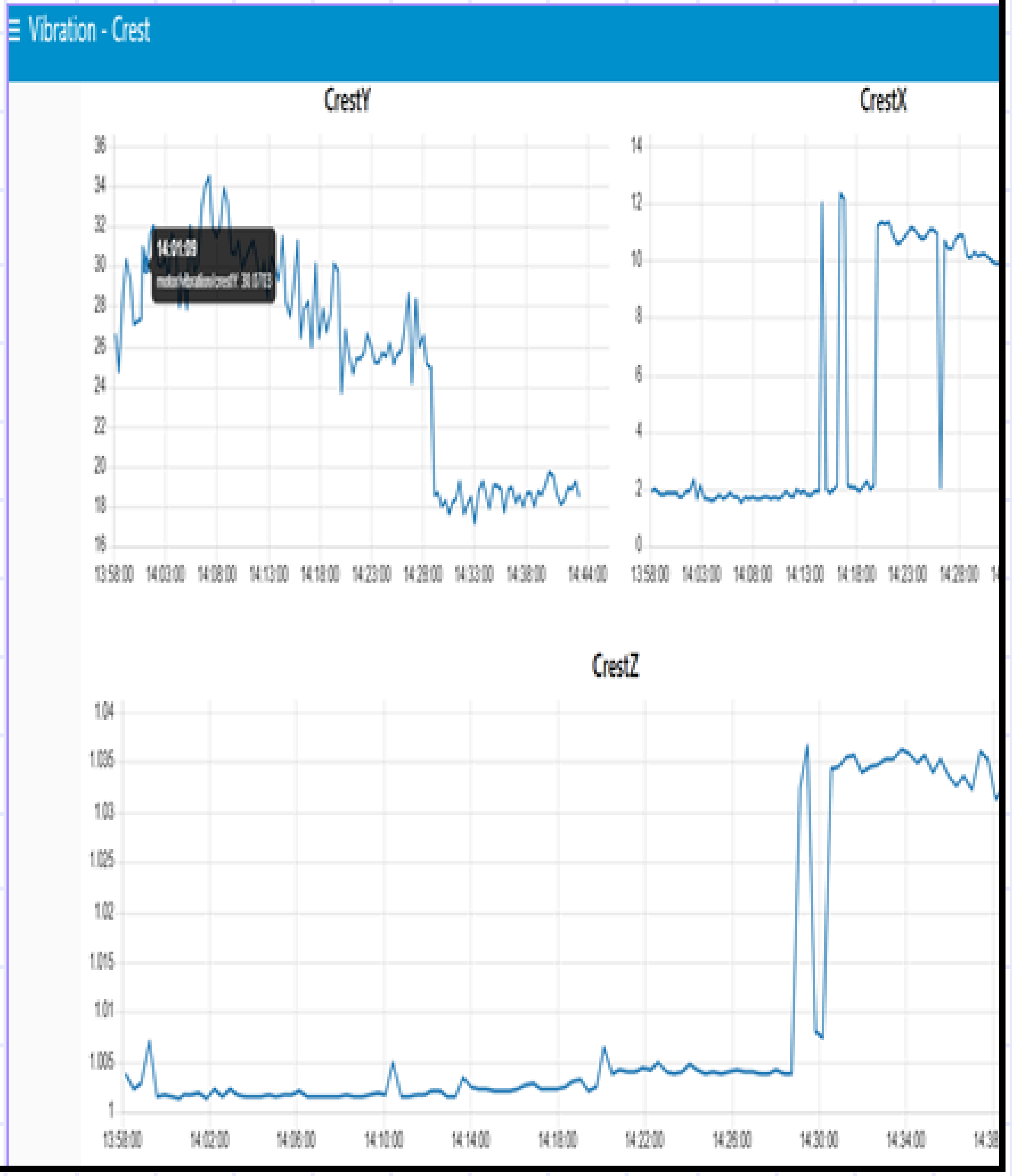
Home — RPM / Current / Temp



Vibration RMS (X, Y, Z)



Vibration Crest Factor (X, Y, Z)



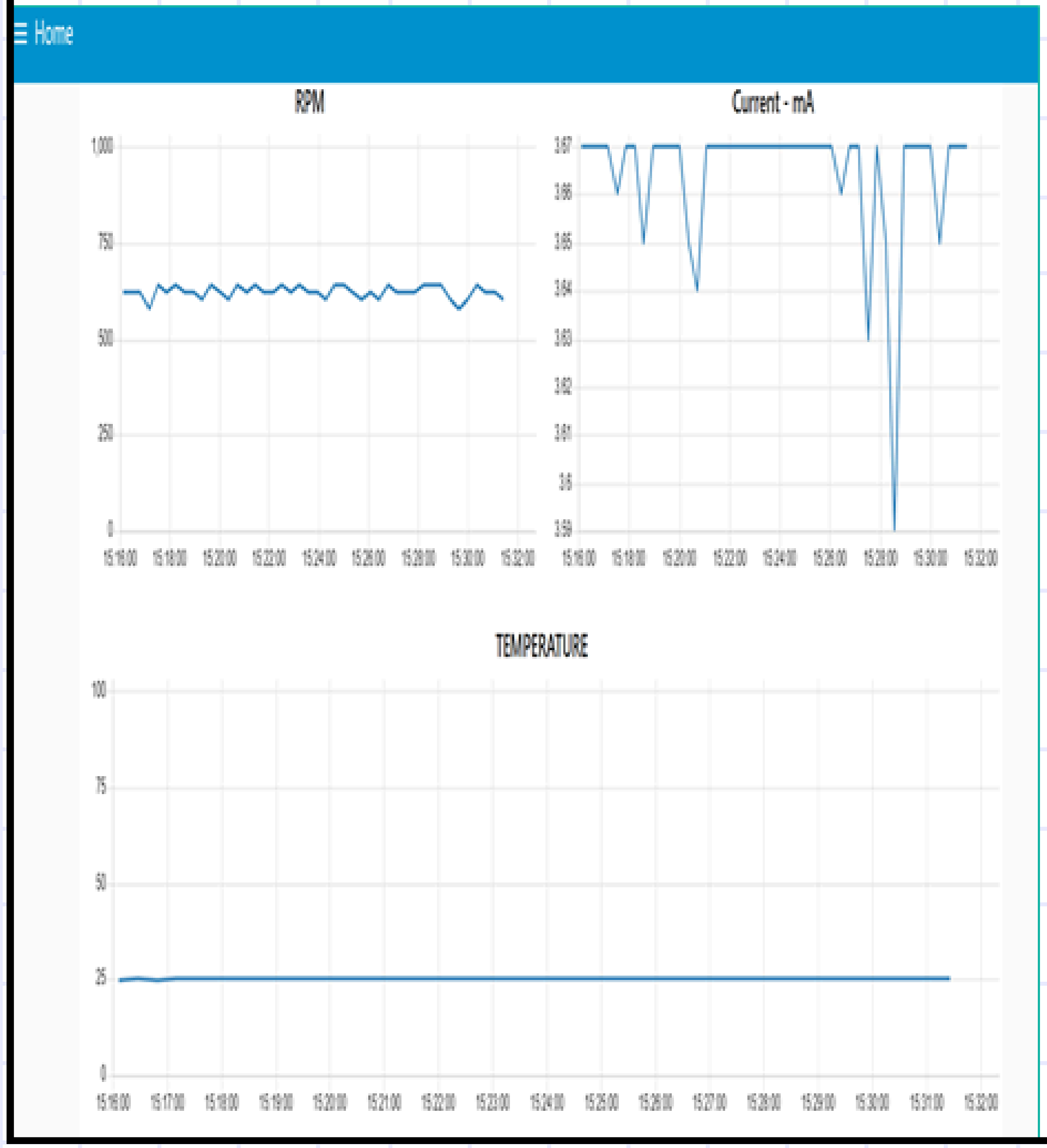
RMS Y is slightly elevated (~0.0085 g) at higher speed – expected and normal. Crest factors are low and stable (~1.4–1.7). RPM stabilises quickly after motor startup. No fault signature present.

DATASET — NO LOOSE SCREW, 1030 RPM

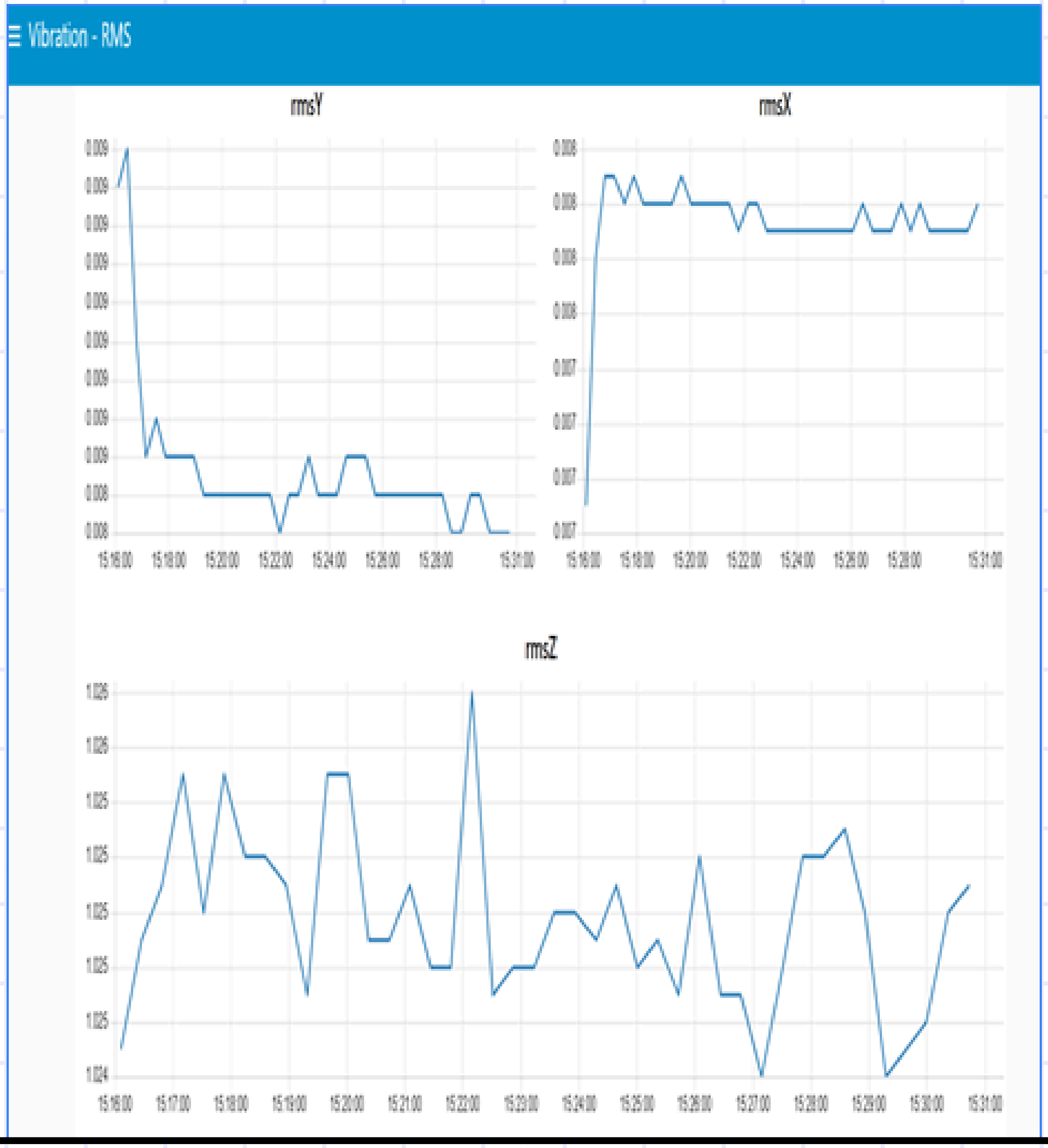
✓ HEALTHY

Healthy condition at higher speed · RPM: ~1030 · Current: ~3.67A · Temp: ~25°C

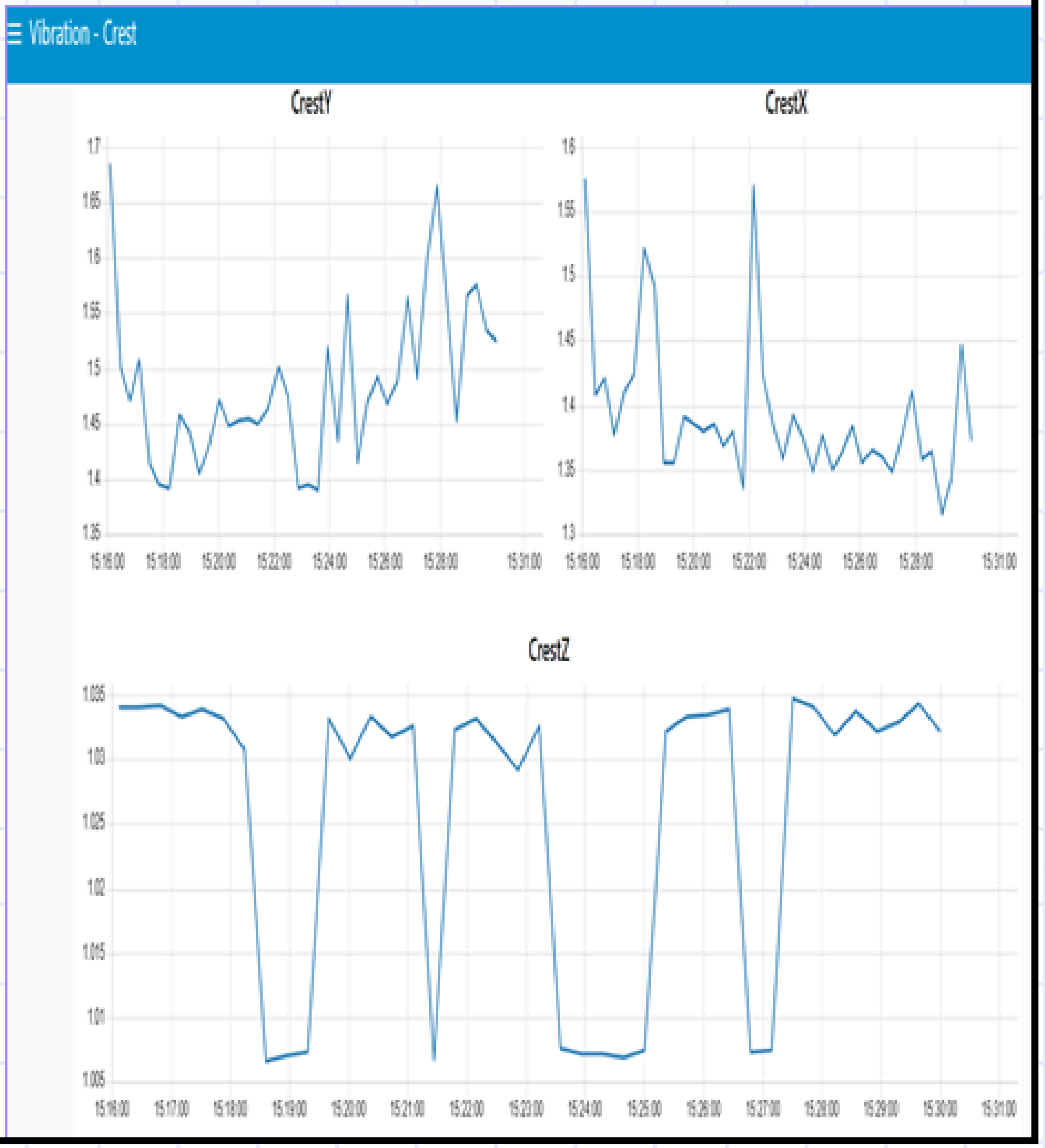
Home — RPM / Current / Temp



Vibration RMS (X, Y, Z)



Vibration Crest Factor (X, Y, Z)



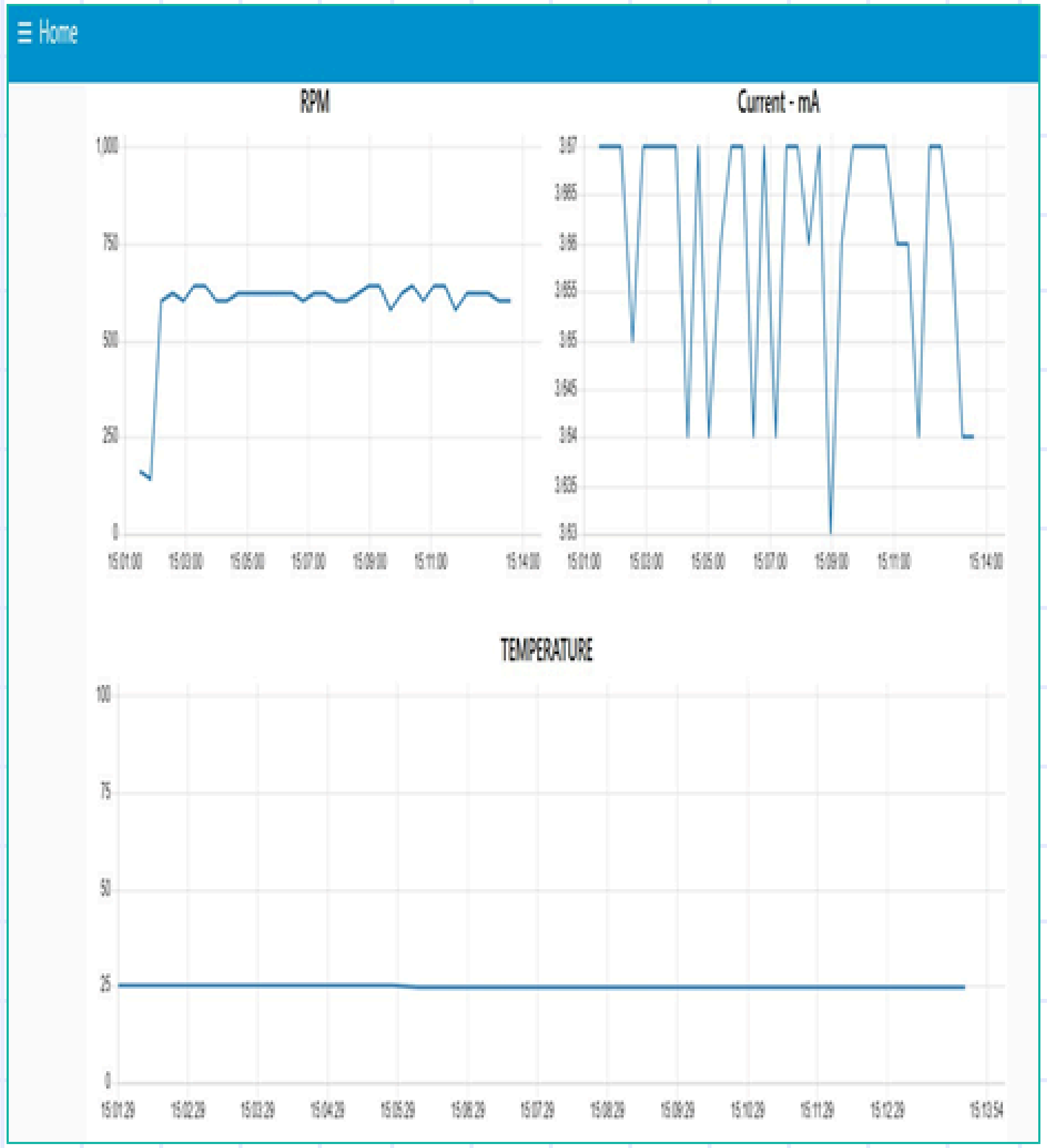
RMS Y is slightly elevated (~0.0085 g) at higher speed — expected and normal. Crest factors are low and stable (~1.4–1.7). RPM stabilises quickly after motor startup. No fault signature present.

DATASET — LOOSE SCREW FAULT, 1030 RPM

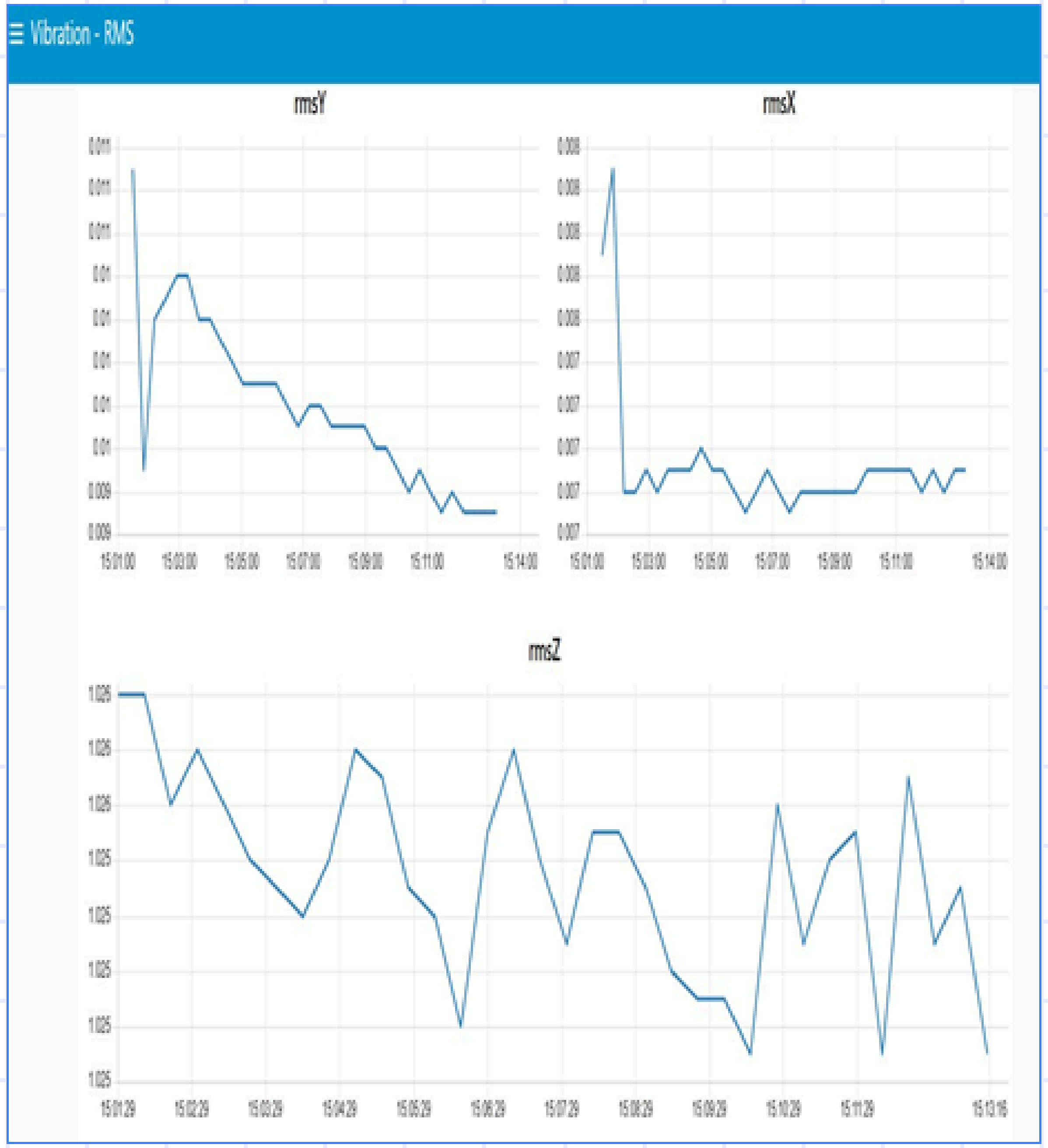
⚠️ LOOSE SCREW

Mechanical fault introduced · RPM: ~1030 · Current: ~3.66A · Temp: ~24.9°C

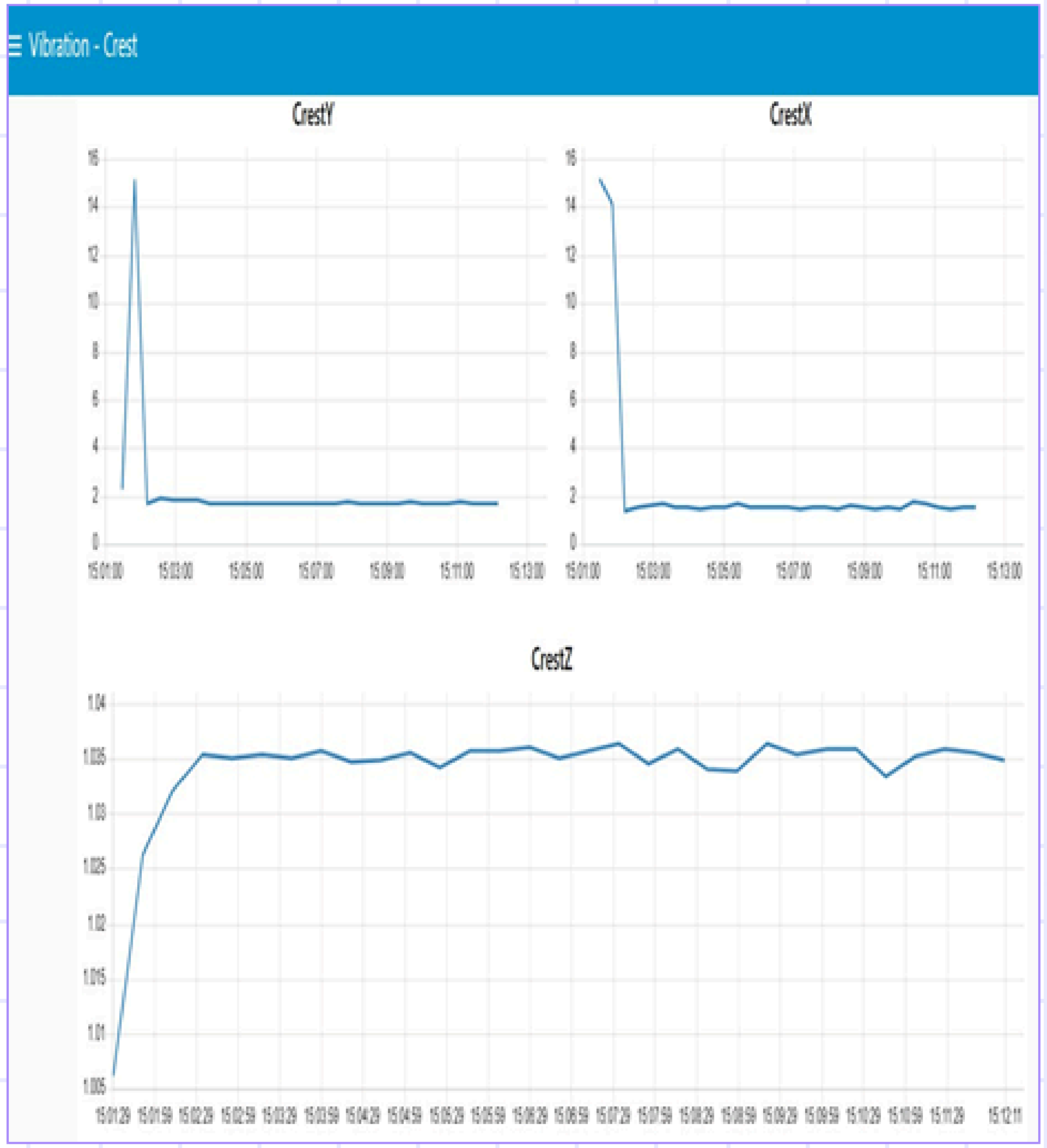
Home — RPM / Current / Temp



Vibration RMS (X, Y, Z)



Vibration Crest Factor (X, Y, Z)



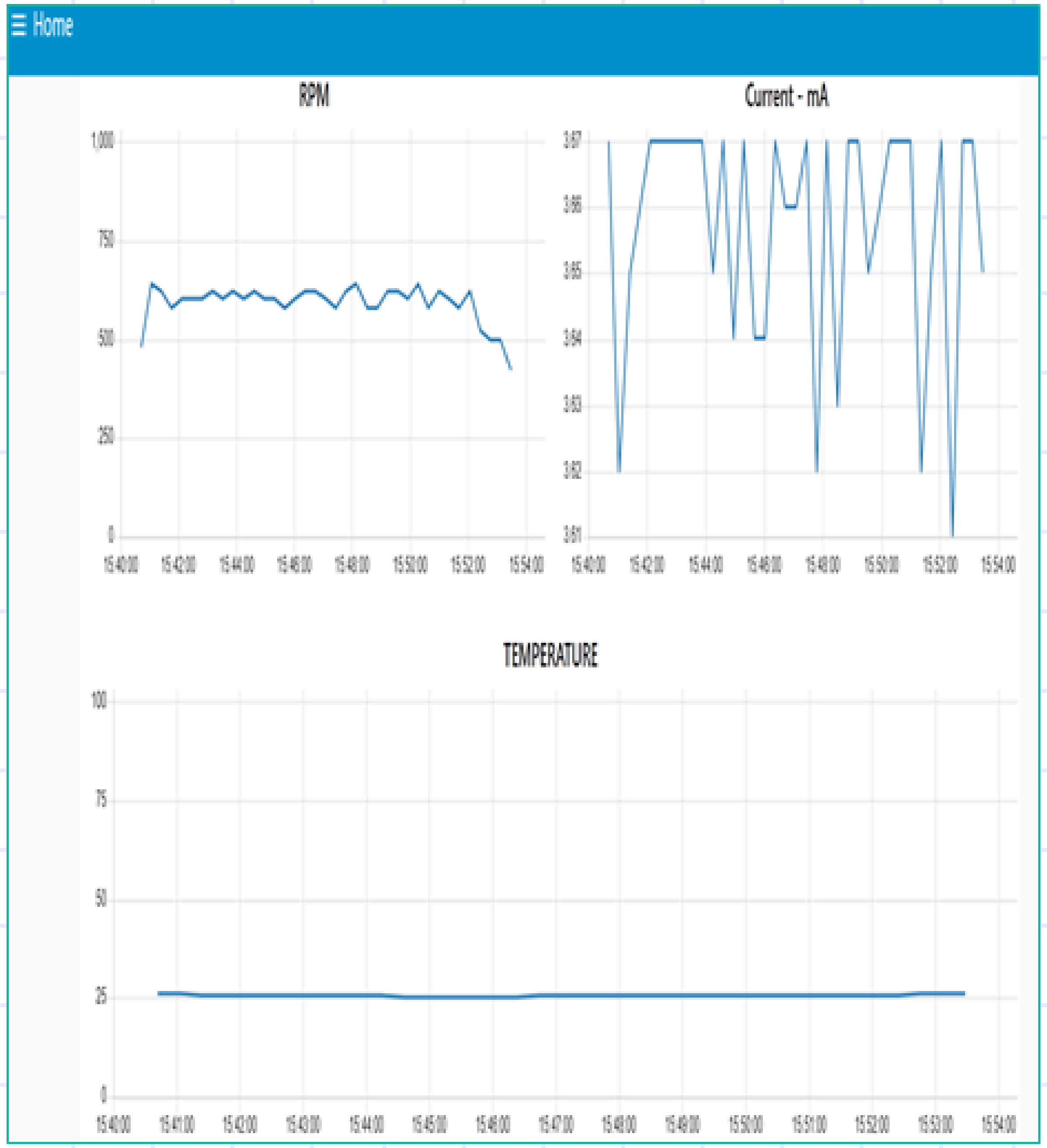
RMS Y clearly elevated (~0.0097–0.010g) — above normal baseline. Vibration settling visible after initial transient. Loose screw introduces periodic mechanical impacts detectable in both RMS and Crest factor.

DATASET — SHAFT MISALIGNMENT, 1030 RPM

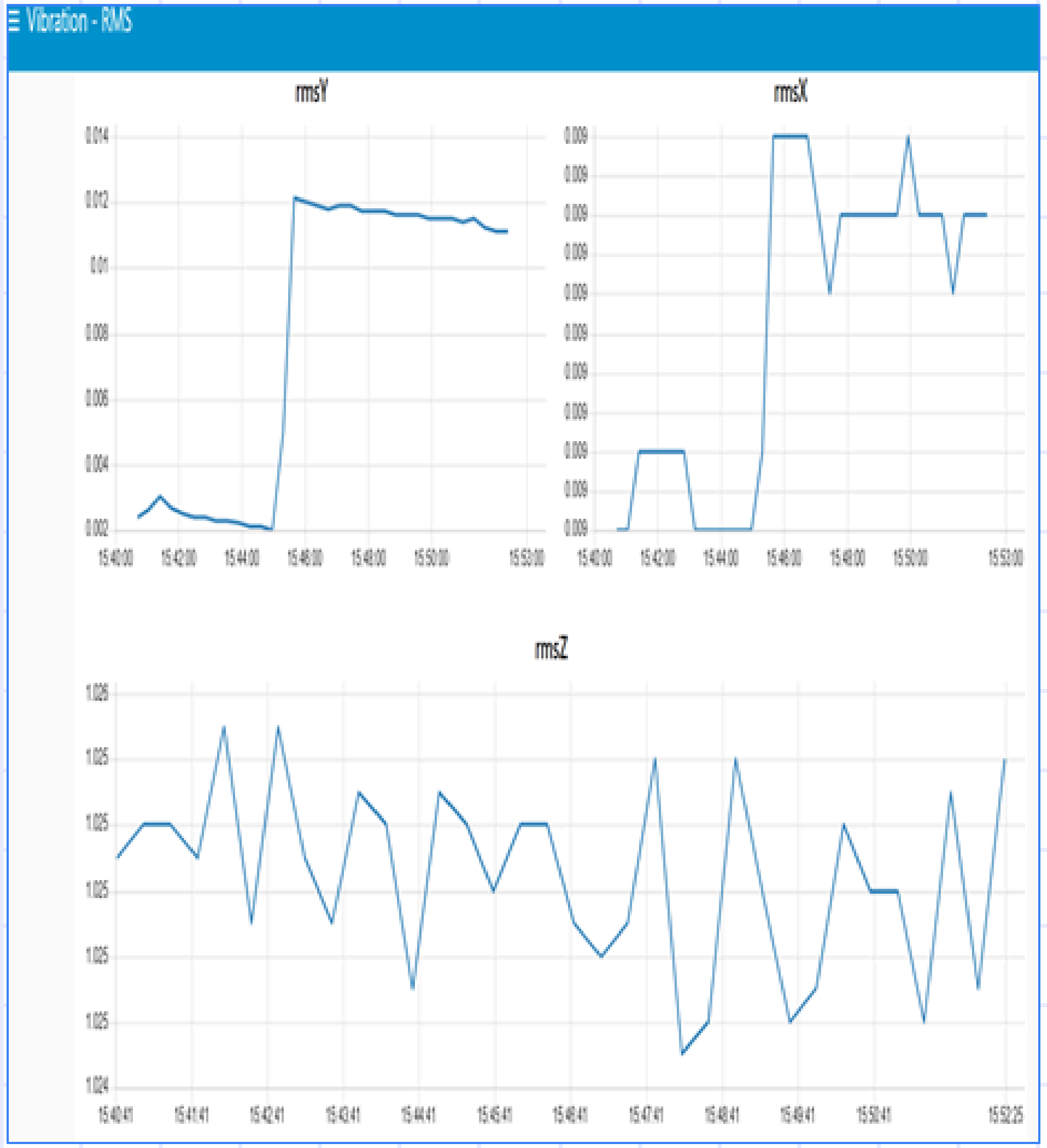
✗ MISALIGNMENT

Misalignment fault introduced · RPM: ~1030 · Current: ~3.66A · Temp: ~25.6°C

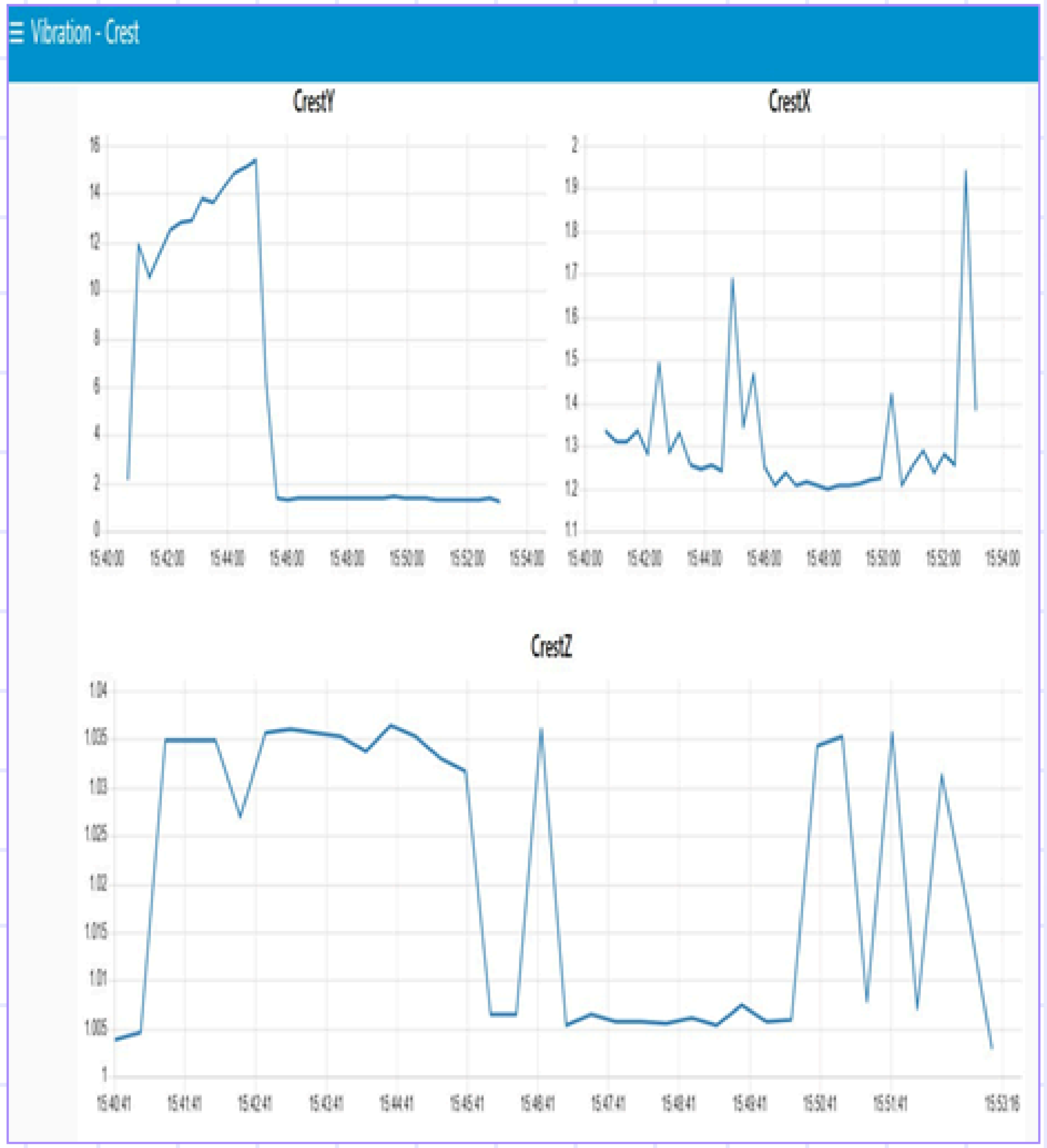
Home — RPM / Current / Temp



Vibration RMS (X, Y, Z)



Vibration Crest Factor (X, Y, Z)



RMS Y large transient spike (0.076–0.084g) on fault start → settles at elevated ~0.011–0.012g. CrestY peaks >15.

MACHINE LEARNING

TIMELINE

INPUT FEATURES	MODELS TRAINED	3 FAULT CLASSES
RMS X / Y / Z	SVM – RBF kernel	Healthy
Crest factor X / Y / Z	Logistic Regression	Misalignment
		SoftFoot

Why only vibration?

Temperature and current showed near-zero variance across fault conditions.

STEP 1

DATA CLEANING

Drop nulls · rename MQTT topics to features · add labels

STEP 2

MERGE DATASETS

Healthy · Misalignment · SoftFoot → shuffled CSV

STEP 3

EDA + SPLIT

Boxplots · PCA · correlation · 80/20 stratified split

STEP 4

TRAIN + EVALUATE

SVM (RBF) · Logistic Regression · GridSearchCV 5-fold

Feature Patterns by Class

Each vibration feature carries distinct discriminative information

rms_x

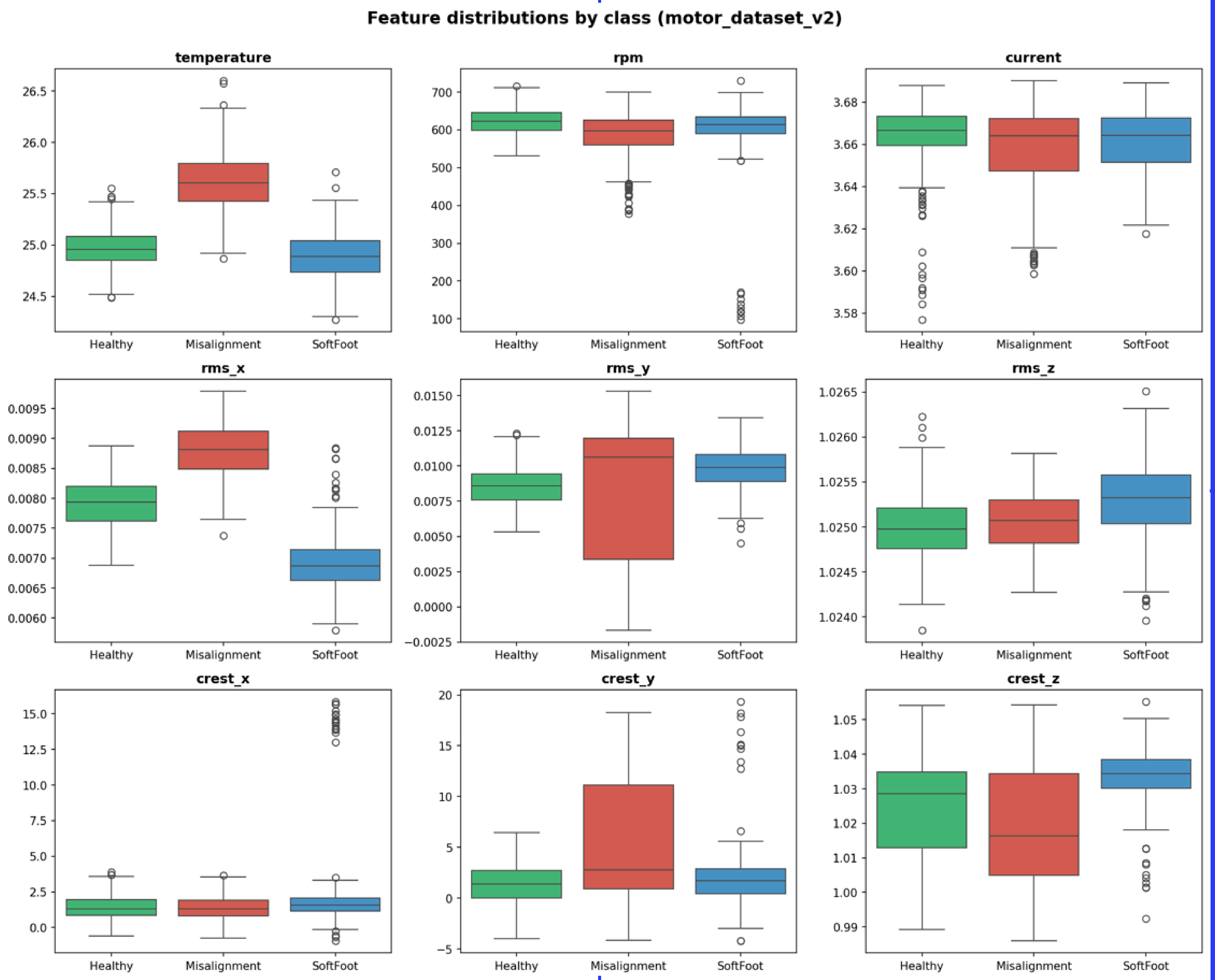
Three classes cleanly separated

crest_y

Misalignment: huge spread, values up to 15

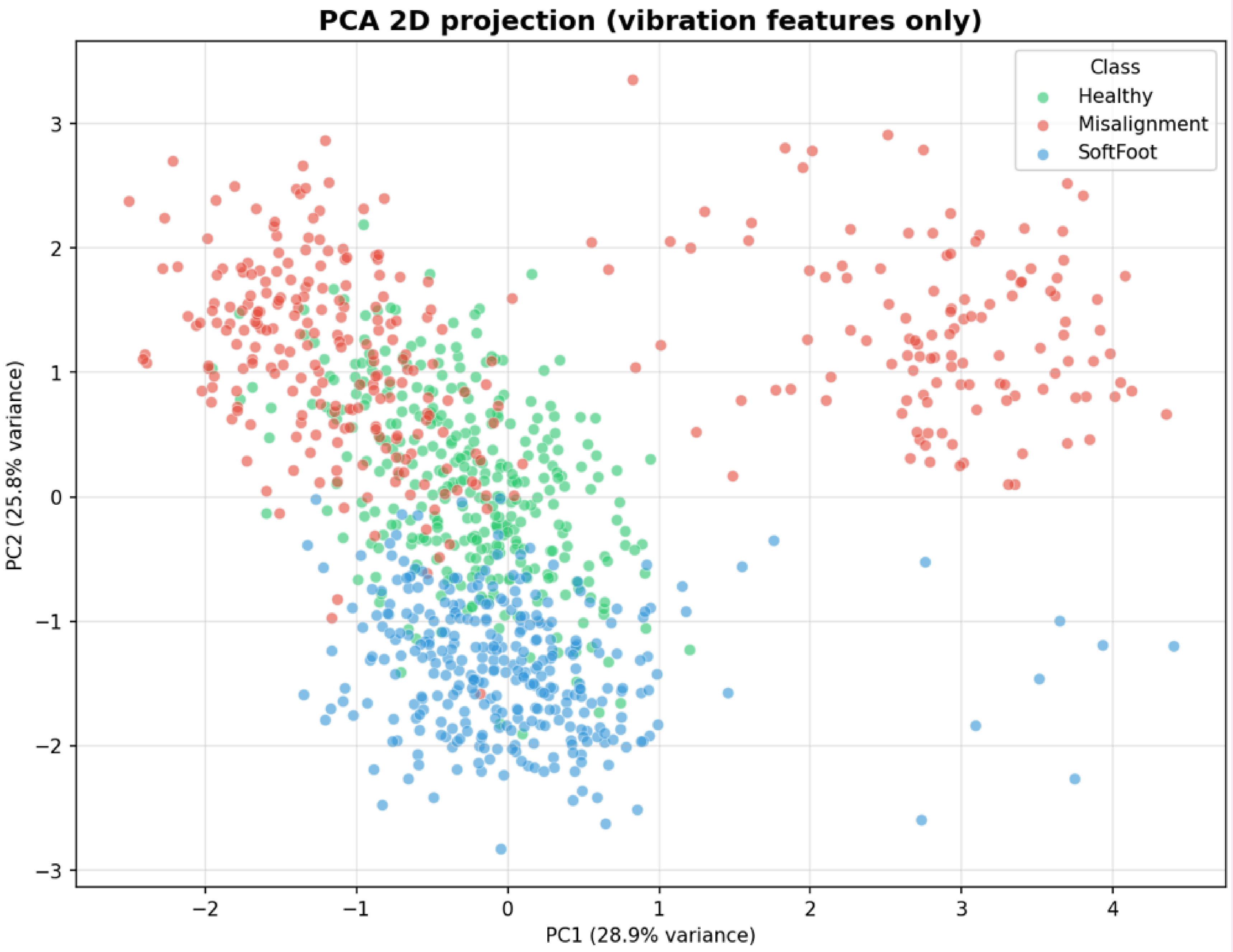
crest_x

SoftFoot: extreme outliers around 14-15



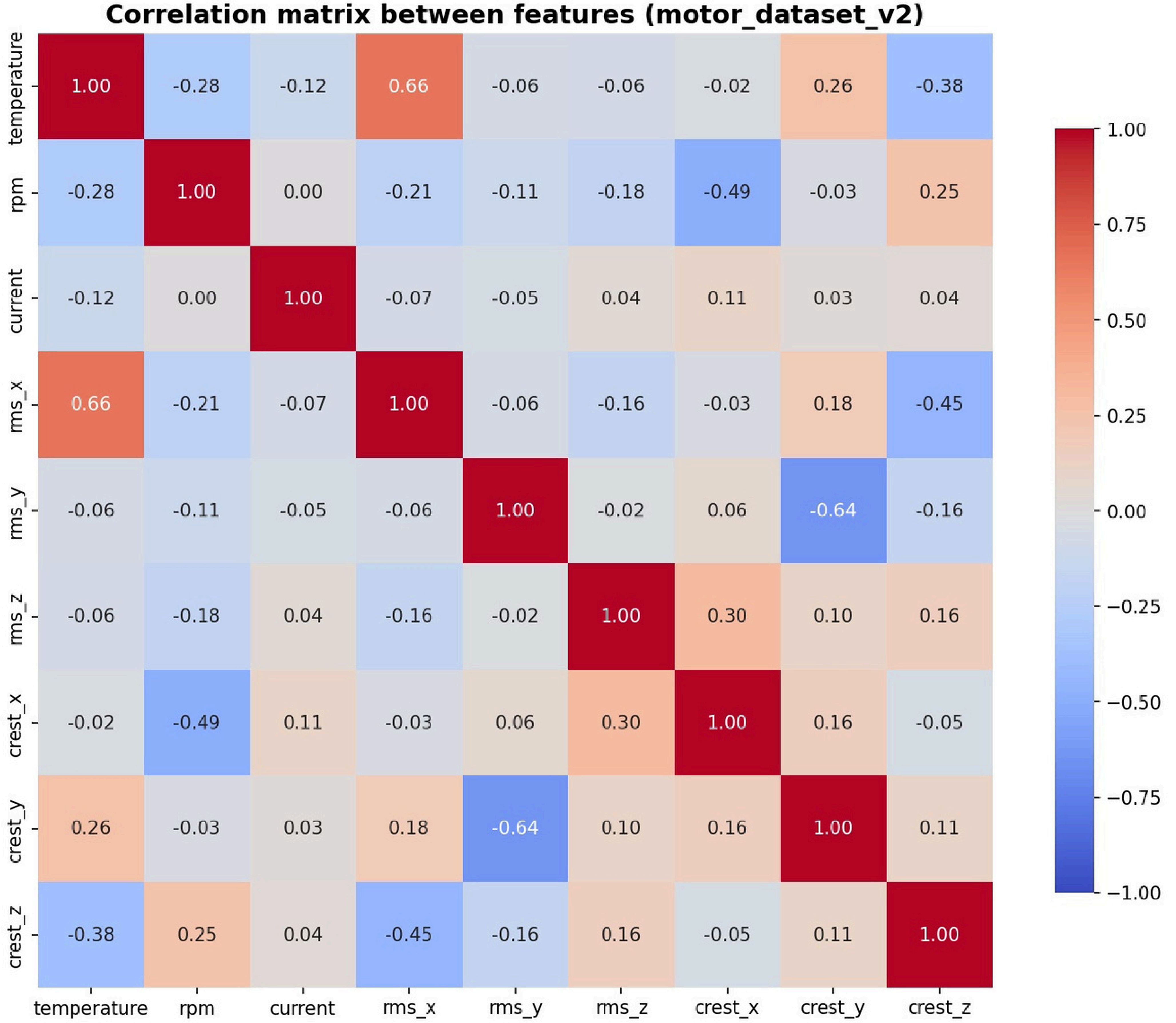
CLASS SEPARATION IN 2D

PCA projection of the six standardised vibration features

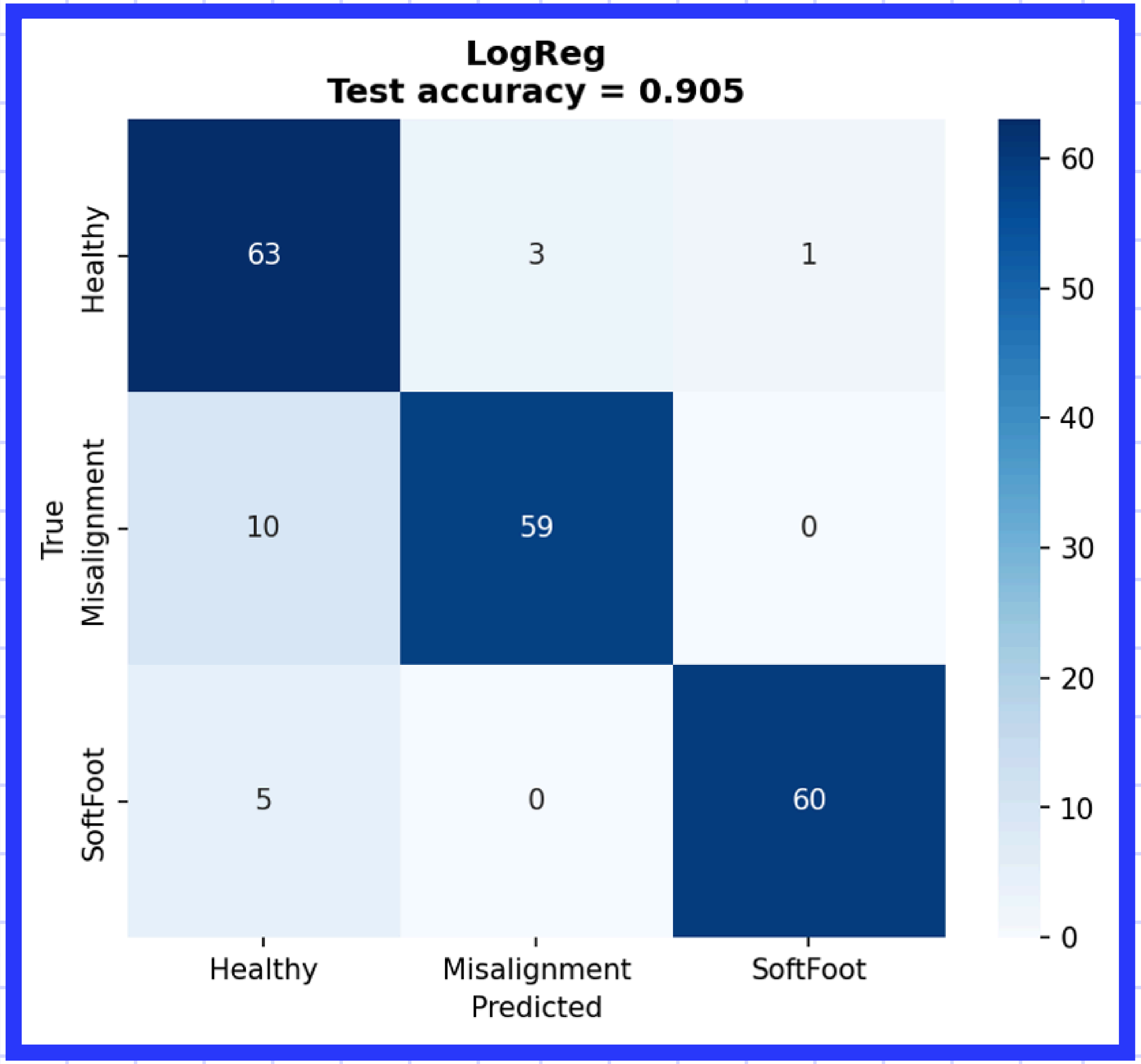


PC1 + PC2 explain 54.7% of the variance

CORRELATION MATRIX



CONFUSION MATRIX



Error Analysis

SVM

8 / 10

errors are faults → healthy

(false negatives for fault detection)

SVM (RBF)

95.0%

Test accuracy | Macro F1 = 0.95

Logistic Regression

90.6%

Test accuracy | Macro F1 = 0.91

Error Analysis

Logistic Regression

15 / 19

errors are faults → healthy

(false negatives for fault detection)

ADVANTAGES

- **LOW COST**

  Shunt resistor + onboard IMU eliminates two expensive sensor modules entirely

- **NO EXTRA WIRING (IMU)**

  LSM6DS3 built into Nano 33 IoT and vibration data gathers with no additional hardware

- **DUAL DASHBOARD**

  Local Node-RED + ThingSpeak cloud gives real-time and historical data access

DISADVANTAGES

- **IMU PLACEMENT**

  Board on breadboard reduces vibration coupling sensitivity vs direct mounting

- **RPM SENSOR NOISE**

  Hall effect RPM sensor was detecting false readings, to overcome that, we used debouncing

- **WIFI DROPOUTS**

  Reconnection logic is implemented in the firmware

FUTURE WORK & CONCLUSION

Mount IMU on Motor Housing:

Direct bearing-housing mounting improves vibration coupling by 3-5X

Train ML on more failure modes

Collect data from more fault categories + Use other ML models

Deploy TinyML On-Device

Run the trained model on Nano 33 IoT for real-time edge inference

Extend to AC Motor

Replace shunt with isolated CT sensor

CONCLUSION:

We successfully implemented a 4-sensor IIoT health monitoring system for a DC motor using an Arduino Nano 33 IoT. Real-time data is streamed via WiFi/MQTT to Node-RED and ThingSpeak. Dataset analysis across 5 test conditions demonstrates clear vibration signatures for fault detection - confirming the system's viability for predictive maintenance

THANK YOU